

Executive Summary:

Early Co-occurrence Network Openness and Cross-Field Concept Spread

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Headline

Concepts whose home-field co-occurrence partners span many communities and arrive through mixed-field papers tend to spread to more disciplines later: pooled partial Spearman $+0.069$ $[+0.038, +0.100]$ over a five-feature popularity baseline ($\text{OPEN}_{\text{home}}$ PSP, DerSimonian–Laird, 6 non-selection bodies, $I^2 = 0$, 6/6 positive). The effect is modest, adds no practical forecasting gain over publication counts, and the temporal interpretation (year-to-year turnover vs static topical breadth) remains unresolved.

Goal

Can early structural signals in a concept’s topic co-occurrence network predict how broadly that concept will later be used across scientific fields? The run screened 53 early network indicators on 12,499 concepts (27,393 adoption episodes across 26 venue fields) from the OpenAlex bulk snapshot (476M works, 2026-09-23), using rarefied field breadth as the primary outcome and a held-out field-group design with hash-sealed specifications.

What was tried

Five iterations tested six classes of candidate predictors:

1. **Iteration 1:** Wide screen of five rivals—citation naturalisation (A_h^*), co-author independent groups (candidate S), co-occurrence diversity (D/F), and gateway-field landing (G)—on a 48-concept DEV panel. All failed the pre-registered rule. The only lead was gateway-field retention ($\Delta\text{AUC} +0.103$ on 80 episodes).
2. **Iteration 2:** Scaled to 12,499 concepts. Gateway retention disconfirmed at scale ($\Delta\text{AUC} = -0.00001$). Retained-field relatedness confirmed for next-field entry ($d_0 = 0.302$).
3. **Iteration 3:** 53-indicator held-out screen: 7 of 10 frozen indicators confirmed across two families (retained frontier and co-occurrence topology). Strongest post-onset: `CONTACT_REACH` ($+0.211$), n_{comm} ($+0.167$), NOV_{res} ($+0.151$).
4. **Iteration 4:** Fresh 2015–2017 cohort ($n = 1,443$) confirmation of the six-component `OPEN` index: $+0.091$ at R2, marginal at R4/R5 (DL includes 0). Breadth decomposition: early contact diversity accounts for $\sim 73\%$ of the breadth gap. Within-concept closure test null on DEV.

5. **Iteration 5:** Frame N vocabulary-free confirmation (636 phrase-born concepts): $\text{OPEN}_{\text{home}}$ +0.117 at R3, +0.086 at R5 (CI includes 0). Cheng reversal confirmed: consistency predicts narrower breadth (-0.069 , $I^2 = 0$, 5/5 negative) despite predicting more volume (+53%/SD; adding current volume removes 98%). Confound check: within-concept permutation absorbs the temporal component (excess +0.008); the signal behaves like static topical non-redundancy.

Key results

1. **Seven confirmed indicators** survive held-out validation for predicting rarefied cross-field breadth, from two families. NOV_{res} (novel co-occurrence partners) carries the home-only signal.
2. **Retained-field relatedness** predicts next-field entry ($d_0 = 0.322$ [0.291, 0.355]), but this is backbone-specific (reverses under min-cp proximity) and the volume-matched contrast is null.
3. **Consistency–breadth reversal:** Cheng et al.’s (2023) ideational consistency is a current-volume proxy (+53% raw, +1.3% conditional). It predicts *narrower* breadth, reversing the sign of their reach claim when the outcome is rarefied breadth rather than raw volume.
4. **Breadth decomposition:** early contact diversity ($\sim 73\%$) dominates; frontier advance contributes near zero. This is an accounting identity, not a causal decomposition.

What failed

- Citation naturalisation A_h^* : null ($\Delta\rho = -0.006$), 0/4 groups positive.
- Gateway-field retention: disconfirmed at scale; it was a domain-specific proxy.
- Co-occurrence diversity D and selectivity F : null on the DEV panel.
- Two trajectory classes: not reproduced by HMM (ARI 0.094); the space is a continuum.
- Within-concept closure test: NOT SUPPORTED (null on DEV, opposite-signed on held-out).
- External recognition (O5): unrelated to breadth; 67% recognised before concept onset.
- $\text{OPEN}_{\text{home}}$ predictive gain: negligible (+0.002 Spearman over B5).
- Four of twenty artifacts failed or were incomplete (Dataset 1, Experiment 2, Experiment 9, Experiment 11).

Open questions

1. **Temporal turnover vs static dispersion:** the within-concept permutation null absorbs most of the signal. Whether year-to-year churn matters beyond static topical breadth could not be adjudicated at typical sample sizes.
2. **Domain heterogeneity:** Life & Environment shows the weakest OPEN signal (PSP +0.071). The reason is unexplained.
3. **Causal mechanism:** association only. Early ego-network openness could reflect intrinsic generality, community diversity, or problem breadth.
4. **Untested rival:** the persistence-filtered RCA density ($D_{\text{rca,persist,k}}$) differs from the tested variant and remains formally untested.